

Alarm Thresholds and Response Matrix — Production Line A

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Threshold hierarchy: CRIT_LO < WARN_LO < NOMINAL < WARN_HI < CRIT_HI.

ST01_FILLING Thresholds

Sensor	Unit	CRIT_LO	WARN_LO	Nominal	WARN_HI	CRIT_HI	CRITICAL Response
TMP	°C	15.0	18.0	22.0 ± 0.4	26.0	30.0	Inspect and notify maintenance
PRS	bar	3.0	3.3	3.8 ± 0.05	4.2	4.8	Inspect and notify maintenance
FLW	L/min	105.0	112.0	120.0 ± 1.5	128.0	135.0	Inspect and notify maintenance

ST02_SEALING Thresholds

Sensor	Unit	CRIT_LO	WARN_LO	Nominal	WARN_HI	CRIT_HI	CRITICAL Response
TMP	°C	165.0	175.0	185.0 ± 1.2	195.0	210.0	E-STOP ST02/ST03/ST04; inspect element
PRS	bar	4.2	4.6	5.2 ± 0.08	5.7	6.2	Inspect and notify maintenance
CUR	A	9.5	11.0	12.4 ± 0.3	13.5	15.0	E-STOP; inspect load; verify ST04-SPD (RULE-ST02-04)

ST03 LABELLING Thresholds

Sensor	Unit	CRIT_LO	WARN_LO	Nominal	WARN_HI	CRIT_HI	CRITICAL Response
SPD	m/s	0.65	0.75	0.85 ± 0.01	0.95	1.05	Inspect and notify maintenance
TEN	N	1.5	1.8	2.2 ± 0.05	2.6	3.0	Inspect and notify maintenance
CNT	pcs/m in	100	110	120 ± 2.0	130	140	E-STOP labelling; inspect label path

ST04_PACKAGING Thresholds

Sensor	Unit	CRIT_LO	WARN_LO	Nominal	WARN_HI	CRIT_HI	CRITICAL Response
VIB	mm/s	0.01	0.02	0.12 ± 0.005	0.2	0.35	E-STOP; inspect main bearing
CUR	A	6.5	7.5	8.7 ± 0.2	9.8	11.0	Inspect and notify maintenance
SPD	m/s	0.85	1.0	1.2 ± 0.02	1.4	1.6	Inspect and notify maintenance

SRV01_SERVERROOM Thresholds

Sensor	Unit	CRIT_LO	WARN_LO	Nominal	WARN_HI	CRIT_HI	CRITICAL Response
TMP	°C	16.0	18.0	21.0 ± 0.3	25.0	28.0	Activate backup cooling; prepare controlled PLC shutdown

HUM	%RH	25.0	30.0	45.0 ± 1.0	60.0	70.0	Inspect and notify maintenance
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WRH01_WAREHOUSE Thresholds

Sensor	Unit	CRIT_LO	WARN_LO	Nominal	WARN_HI	CRIT_HI	CRITICAL Response
TMP	°C	-1.0	0.5	4.0 ± 0.5	8.0	12.0	Activate backup refrigeration; notify QC
HUM	%RH	55.0	60.0	75.0 ± 1.5	85.0	90.0	Inspect and notify maintenance

CHM01_CHEMICALSTORAGE Thresholds

Sensor	Unit	CRIT_LO	WARN_LO	Nominal	WARN_HI	CRIT_HI	CRITICAL Response
TMP	°C	5.0	10.0	15.0 ± 0.3	20.0	25.0	Activate emergency ventilation; notify safety officer immediately
HUM	%RH	15.0	20.0	35.0 ± 1.0	50.0	60.0	Activate desiccant system; inspect storage containers for leaks

RND01_RDLAB Thresholds

Sensor	Unit	CRIT_LO	WARN_LO	Nominal	WARN_HI	CRIT_HI	CRITICAL Response
TMP	°C	15.0	18.0	22.0 ± 0.2	26.0	30.0	Inspect and notify maintenance
HUM	%RH	25.0	35.0	50.0 ± 1.0	65.0	75.0	Suspend experiments; inspect HVAC system

CAF01_CAFETERIA Thresholds

Sensor	Unit	CRIT_LO	WARN_LO	Nominal	WARN_HI	CRIT_HI	CRITICAL Response
TMP	°C	14.0	17.0	21.0 ± 0.4	26.0	30.0	Inspect and notify maintenance
HUM	%RH	25.0	35.0	55.0 ± 1.5	70.0	80.0	Inspect and notify maintenance

Anomaly Type Definitions for KG Mapping

Type	Definition	Detection Criterion	Ontology Class
SPIKE	Transient deviation >3σ, <5 min	Single reading >CRIT	AnomalyEvent::Spike
DRIFT	Monotonic deviation >15 min	Slope >0.1σ/min for 5+ readings	AnomalyEvent::Drift
STUCK	Unchanged >10 consecutive readings	std(last 10)<0.001×nominal	AnomalyEvent::StuckSensor
OUT_OF_RANGE	Sustained WARN violation >10 min	>WARN for 20+ readings	AnomalyEvent::OutOfRange
CORRELATED	Causally linked to another station	Onset within 90 min of cause	AnomalyEvent::Correlated